
Reading Positions Off a Genotype–Phenotype Model: A Negative Result on Attribution Reliability in a Multiplexed Splicing Assay

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Abstract

1 Reading the top attributed positions off a trained sequence–activity model is stan-
2 dard practice for guiding RNA therapeutic design [14], and that community has ar-
3 gued such claims are validated anecdotally [22]. We set out to validate the practice
4 on a real multiplexed splicing assay and report that it fails. Models whose held-
5 out accuracy is statistically indistinguishable assign different top-3 positions for
6 43% of held-out sequences at the best-fitting cell in our grid, rising to 94% at the
7 worst. The failure is therefore not confined to poorly-fitting models; no cell in the
8 grid escapes it. Diagnosing why, we find three compounding causes. The fitting
9 objective does not identify the attribution, so the choice among equally-accurate
10 models is made by the initialization seed. Attribution mass is concentrated — a
11 median 98% of a sequence’s total in 3 of 9 positions at the best-fitting cell — and
12 we show in closed form that under such concentration a Spearman correlation of
13 0.71 is what *perfect* agreement about which positions matter already produces, so
14 the rank correlations normally used to check agreement cannot distinguish success
15 from failure. And the criterion the field selects on does not constrain the function:
16 tied models differ pointwise on the scale of the assay’s own Poisson observation
17 noise, a median prediction difference of 0.5 to 2.5 times the noise standard devia-
18 tion, so one or two held-out measurements separate them as functions even though
19 their aggregate accuracy is indistinguishable — the data could choose, and the ac-
20 curacy criterion does not. We also report two ways our own validation nearly hid
21 the result: a control that read as passing because the refit began at its own target,
22 and a negative reported from a search that examined zero candidates. We report
23 the conditions under which the failure is worst rather than any under which the
24 practice is safe, since we found none, and a reporting checklist.

25 1 Problem

26 **Biological task.** Given a short RNA sequence and a measured activity, identify *which positions*
27 *determine that activity*. In splice-site and siRNA design this is not a secondary output; it is the
28 object that guides the next round of synthesis. A designer who reads positions 2, 5 and 6 as critical
29 will vary those and hold the rest fixed.

30 **Data modality and setting.** A multiplexed assay of variant effect: each sequence is assayed in
31 a pooled library and read out as a ratio of two sequencing count pools. We use the BRCA2 exon
32 17–19 5’ splice-site assay of Wong et al. [20], which quantifies $\sim 30,483$ nine-nucleotide sites with
33 $\log_{10}$ percent-spliced-in readouts and retains per-sequence read counts. We subsample $n = 4000$
34 per seed, split 3000 fit / 1000 held out by a fixed seeded permutation recorded with each run.

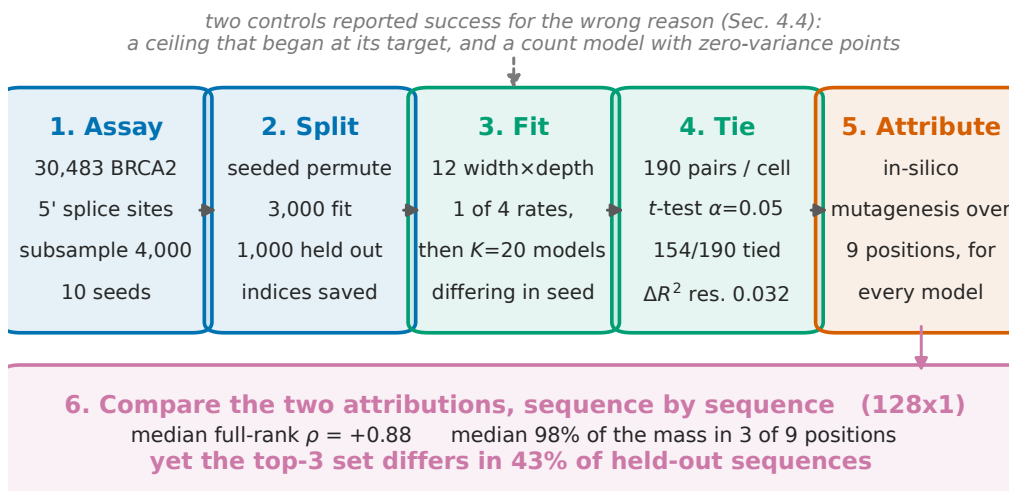


Figure 1: **The pipeline, and where the disagreement appears.** Every count is read from the committed run behind this paper. Models are trained per width $\times$ depth cell differing *only* in initialization seed, then filtered to the pairs a paired t -test on held-out squared errors cannot separate at $\alpha = 0.05$; the surviving pairs are compared by in-silico mutagenesis, sequence by sequence. At the best-fitting cell the survivors agree well on the full-position ranking and concentrate almost all their mass in three positions, yet they disagree about *which* three for nearly half of held-out sequences. The two controls named above the pipeline both reported success without testing what they were meant to test (Section 4, Table 3).

35 **Assumption under test.** That the top- k attributed positions recovered from a well-fitting model
 36 are a property of the sequence–activity relationship rather than of the particular model that happened
 37 to be trained.

38 **Target metric.** Agreement of the top- k attributed position *set* between models that *predictive accu-*
 39 *racy* cannot distinguish, measured per-sequence rather than averaged over the library. The qualifier
 40 matters and is the subject of this paper: accuracy-matching is a statement about one aggregate scalar,
 41 not about the functions being close. We use $k = 3$ throughout, justified in Section 4 by the measured
 42 concentration of attribution mass and by Proposition 1 in Appendix A.

43 **Desired outcome.** For the practice to be sound, accuracy-matched models should agree on the top-
 44 k set for nearly all sequences. A designer should not get a different answer because a different seed
 45 was used. If they disagree, either the assay cannot tell the models apart, or it can and the selection
 46 procedure is discarding information the data already holds. We find the second.

47 2 Proposed approach

48 The approach under investigation is standard practice [2, 19], applied in genomics with in-silico
 49 mutagenesis since [21] and still standard [3, 11, 17], not a method we invented: fit a genotype–
 50 phenotype model to the assay, compute per-position attributions by in-silico mutagenesis, and read
 51 the top-ranked positions as the determinants.

52 **Core mechanism.** A model f_θ is fit by minimizing predictive loss on the assay. Attribution scores
 53 each position by the mean absolute change in prediction under substitution, over the real four-letter
 54 alphabet:

$$A_j(x) = \frac{1}{3} \sum_{b \neq x_j} |f_\theta(x^{j \rightarrow b}) - f_\theta(x)|. \quad (1)$$

cell	held-out R^2	tied pairs	resolution (ΔR^2)	ρ p10 / median	frac $\rho < 0.5$
16 $\times$ 1	0.6011	162/190	0.0699	+0.217 / +0.650	30%
16 $\times$ 2	0.5591	158/190	0.1059	-0.167 / +0.417	58%
16 $\times$ 3	0.5349	134/190	0.1161	-0.300 / +0.283	69%
128 $\times$ 1	0.6372	154/190	0.0318	+0.700 / +0.883	2%
128 $\times$ 2	0.5836	144/190	0.0461	+0.400 / +0.750	16%
128 $\times$ 3	0.4761	95/190	0.0937	-0.267 / +0.317	67%

Table 1: **Models the held-out data cannot tell apart still rank a sequence’s positions differently.**

Accuracy-tied pairs and their attribution agreement. A pair is *tied* when a paired two-sided t -test on per-point held-out squared errors does not reject at $\alpha = 0.05$; *resolution* is the smallest held-out R^2 gap that test could have detected at 80% power, so a tie is a failure to reject at a stated resolution rather than a claim of equality. Per-instance ρ is a Spearman correlation between two models’ 9-position attribution vectors for a single held-out sequence, pooled over sequences and over tied pairs. **Rows are the two extreme widths at each depth**, which bracket both axes the argument turns on; all twelve cells, with bootstrap intervals, are in Table 5.

55 Positions are ranked by A_j and the top k reported.

56 **Preconditions.** (i) The model fits the assay well enough that its sensitivities are meaningful. (ii)
57 The attribution is faithful to the model, in the sense that it reflects the model’s own dependence on
58 each position [10]. (iii) Implicitly, that the model is *the* model — that the fitting procedure selects a
59 function determined by the data.

60 **Hypothesis under which it works.** If the data determines f up to predictive equivalence, and
61 attribution is faithful to f , then the top- k set is determined by the data. Precondition (iii) is the one
62 we test, and it is the one that is almost never stated.

63 **Models and protocol.** Feedforward genotype–phenotype maps over a grid of widths
64 {16, 32, 64, 128} and depths {1, 2, 3}, with the per-cell learning rate chosen by pilot on the fit split.
65 Within a cell we train $K = 20$ models differing *only* in initialization seed: same architecture, same
66 split, same budget, same learning rate. All reported quantities use at least 5 seeds with bootstrap in-
67 tervals; enumerations are exact and reported as exact. Claims and thresholds were pre-registered and
68 committed before any run existed, and their single-commit history predates the earliest run directory
69 (Appendix C). Total compute is on the order of GPU-days on a single mid-range machine.

70 3 Observed outcome

71 Models that predictive accuracy cannot separate assign different top-3 positions for 43% to 94%
72 of held-out sequences. They are not, however, hard to tell apart: fewer than three held-out mea-
73 surements reject that any such pair is the same function. This mirrors the disagreement problem
74 documented for general-purpose explainers [12], and joins results showing genomic sequence mod-
75 els can score well while generalizing poorly [16]. **The practice fails by discarding recoverable**
76 **information.** Selection interrogates one aggregate scalar, and two models can sit inside its resolu-
77 tion while predicting pointwise differences on the scale of the assay’s noise. Whichever model the
78 seed produced is then read for attributions as though the data had chosen it.

79 **Accuracy matching.** We filter to pairs whose held-out performance a paired t -test on per-point
80 squared errors cannot separate at $\alpha = 0.05$. This is a failure to reject, not a demonstration of
81 equivalence, so we report the filter’s resolution alongside it: the minimum detectable held-out R^2
82 difference at 80% power given $n_{\text{held-out}} = 1000$. Of 2280 pairs, 1610 are not separable at this
83 resolution.

84 **It is not an artifact of weak models.** Held-out R^2 does correlate with agreement across cells
85 (Spearman +0.650 against the minimum ρ), so we state the result at the *best*-fitting cell rather than
86 the worst. At width 128, depth 1 — held-out $R^2 = 0.637$, the highest in the grid — 154 of 190

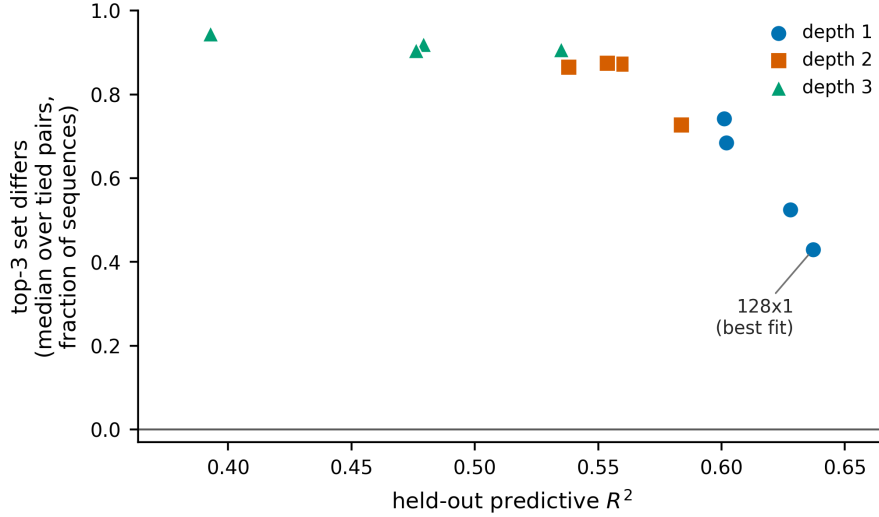


Figure 2: **The failure is not an artifact of poorly-fitting models.** Each point is one architecture cell: held-out predictive R^2 against top-3 attributed-set divergence between accuracy-matched models. **The plotted quantity is the median, over accuracy-tied pairs, of that pair’s fraction of held-out sequences whose top-3 sets differ** — a median over pairs of a per-pair fraction, not a single fraction pooled over sequences; the same definition is used in Tables 2 and 5. The best-fitting cell still diverges on 43% of sequences, so the failure cannot be dismissed as a symptom of underfitting.

87 pairs are accuracy-tied, and among them 37% rank positions at ρ below 0.9, 15% below 0.8, and 6%
88 below 0.7, reaching +0.350 (Figure 2).

89 **The top- k set, which is what gets used.** The top-3 set differs for 73% and 90% of held-out
90 sequences at width 128, depths 2 and 3; restricted to sequences where the concentration diagnostic
91 (Section 4) says the comparison is meaningful, 71% and 80%. The conditioned figure carries little
92 weight at depth 3, where only 11% of sequences at width 128 and none at width 64 have a well-
93 defined top-3 under both models.

94 **Dominant position, or the order beneath it?** A shuffle of ranks 2 and 3 is a far weaker finding
95 than a change of leader, and the top- k Jaccard cannot separate them: it is over unordered sets, so two
96 models can select the same three positions and still disagree about which dominates. We therefore
97 report the strongest position separately (Table 2). **The two models pick a different strongest**
98 **position for 18% of sequences at width 128 depth 1, rising to 68% at depth 3** — at the best cell,
99 one sequence in five whose dominant position, carrying a median 71% of the mass on its own, is not
100 the same position under a model tied on accuracy. Pooling the 203,200 sequence-pair observations
101 at that cell rather than taking a median over pairs gives 19.7% and a 44.2% set disagreement, which
102 decomposes as 35.0 points agreeing on the leader but differing at ranks 2–3, 9.2 differing on both,
103 and 10.5 selecting the same three positions but a different leader. **So at the best-fitting cell most of**
104 **the top-3 disagreement is about ranks 2 and 3, and we say so rather than claim the stronger**
105 **result**; at depth 3 the balance inverts, with 63.9 of the 91.0 points of set disagreement also involving
106 a different leader. These are accuracy-tied pairs of *independently trained* models, the population
107 described in Section 2. The corresponding figures for a fit against its own reparameterized twin are
108 much smaller — 46%, 47%, 34% and 21% — and are reported separately in Appendix 5; the two
109 populations must not be quoted interchangeably.

110 **How cheaply the data adjudicates.** Proposition 3 gives the measurements needed to reject that
111 two accuracy-tied models are the same function, under a noise scale derived from the assay’s own
112 read counts (Proposition 2). The requirement is a median of 0.3 to 2.1 in every cell, four orders of
113 magnitude below the 30,483-sequence library (Figure 3); the median pointwise prediction difference
114 runs 0.5 to 2.5 times the observation noise. One held-out measurement would settle which model

cell	top-3 mass	effective positions	top-1 differs	top-3 exact match	mean Jaccard
16×1	0.926	3.03	45%	26%	0.57
16×2	0.819	4.44	60%	13%	0.43
16×3	0.768	5.11	68%	9%	0.37
128×1	0.976	2.20	18%	57%	0.78
128×2	0.918	3.16	36%	27%	0.60
128×3	0.728	5.67	68%	10%	0.38

Table 2: **The disagreement is about which positions carry the mass, not about the ordering of a negligible tail.** Top-3 set agreement and attribution concentration. *top-3 mass* is the median fraction of a sequence’s total $|\Delta|$ in its three strongest positions and *effective positions* is exp of the entropy of the normalized magnitudes, where 9 means all positions contribute equally. Jaccard on 3-element sets takes only $\{0, 0.2, 0.5, 1\}$, so its median is near-useless as a summary and the *mean* is reported, alongside the fraction of sequences whose top-3 sets match exactly. **top-1 differs** is the fraction of sequences on which the two models’ single *strongest* position is a different position — a stronger form of disagreement than a reordering beneath a shared leader, and not implied by the top-3 columns, which are over unordered sets. Rows are the same bracketing subset as Table 1; all twelve cells are in Table 5.

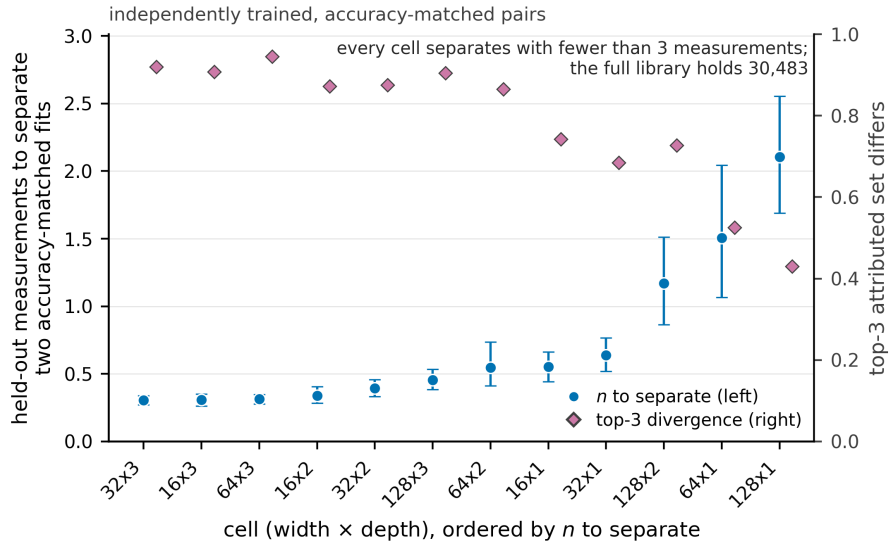


Figure 3: **Accuracy-tied models are cheap to tell apart as functions, yet disagree about which positions matter.** Held-out measurements required to reject that two *independently trained* accuracy-matched models are the same function (Proposition 3), under the assay’s own Poisson count-noise model, with the top-3 set divergence on the right axis. Every cell needs fewer than three measurements, against a library of 30,483, so the reference line is off scale and is stated in the panel rather than drawn. The companion figure for a fit against its own reparameterized twin, where the requirement spans nine orders of magnitude, is Appendix 5.

115 is right, and selection never takes it. The complementary case — a fit against its own *reparameter-*
116 *ized twin*, needing up to 8.8×10^7 measurements yet diverging on top-3 anyway — is a different
117 population, reported in Appendix G.

118 4 Reason for failure

119 4.1 Cause 1: the fitting objective does not identify the attribution

120 Predictive loss constrains f on the observed points. It places no constraint on which of the many
121 equally-accurate functions is selected, and that selection is made by the initialization seed (Re-
122 mark 1). This is the Rashomon effect [5, 7, 9, 13, 18] instantiated on a real assay, and the same

underspecification documented across ML pipelines [6]. What is new here is not that multiplicity exists but its magnitude *at the level of the top- k set*, and — the point this paper turns on — that the multiplicity is *not* forced by the data.

The selection criterion is too coarse, which is a different problem from too little data. The accuracy filter cannot resolve a held-out R^2 gap below 0.032 even at the best-fitting cell, yet two models inside that gap are separable as functions by fewer than three measurements (Section 3). The assay distinguishes them cheaply and the procedure that picked one never asked: the information is in the data and the selection step discards it. Had the models been indistinguishable the remedy would be a larger assay; because they are not, it is to compare functions rather than aggregate error (Section 4.6), or to prefer a model interpretable by construction [15].

The consequence is that precondition (iii) of Section 2 is false, and it is the precondition nobody states. Faithfulness of the attribution to the model (precondition ii) does not help: a perfectly faithful attribution of an arbitrarily-selected model is an arbitrary answer.

4.2 Cause 2: attribution is concentrated, so the usual check cannot detect the failure

Attribution is not spread evenly across positions, and how unevenly depends strongly on the cell. The median fraction of a sequence’s $|\Delta|$ in its three strongest positions runs from 54% (width 64 depth 3) to 98% (width 128 depth 1), grid median 86%; effective positions run 7.80 to 2.20 of 9, grid median 3.92 (Table 2, Proposition 4). Figure 4a shows the most concentrated cell, width 128 depth 1: a median 97% in three positions, 0.71 at the single strongest. Quote a per-cell value or a grid median deliberately — they differ by more than ten points here — and match the population, since the twin comparison gives a grid median of 82% and 4.38.

This is not merely a nuisance; it makes the standard check uninformative, and we can say by how much. Attribution is already known to be fragile to small perturbations [8], and there are limits on what any attribution method can certify about a model [4]; the point here is narrower and empirical — the statistic normally used to *check* agreement cannot see the failure. Proposition 1 shows that if two models agree *exactly* on the top 3 of 9 positions and order the remaining 6 at random, the expected Spearman correlation is 0.708. A reported ρ of 0.7 is therefore consistent with perfect agreement about which positions matter, and also with substantial disagreement. The statistic cannot distinguish the two.

Diagnostic. Report the top- k position *set* overlap and the concentration alongside it. Under that diagnostic, ten of twelve cells show genuine disagreement about which positions matter, and it does so in every cell of the grid. Exact top-3 agreement runs 26–57% at depth 1, 12–27% at depth 2 and 6–10% at depth 3. The best cell in the grid, width 128 depth 1, still assigns a different top-3 set for 43% of held-out sequences. We had expected a low-capacity exemption and there is none: the smallest models are among the *worst*, not the best. See Section 4.5.

4.3 Cause 3: the noise model, and a silent failure in ours

Asking how much data would separate two models requires a noise scale, and this assay supplies one rather than needing an assumption. The phenotype is affine in $\log_{10}$ of a count ratio: regressing y on $\log_{10}(\text{ex}/\text{tot})$ over the library gives slope 0.933 with $r = 0.990$. Those are two independent count pools, not a proportion — ex exceeds tot in 4.9% of rows, by up to $20.9\times$ — so a binomial model is wrong here. Proposition 2 gives the Poisson log-ratio scale.

Our first noise model failed silently, in the direction that hides the result. Under a binomial form, a near-saturated sequence receives $\sigma \rightarrow 0$; since the separation test of Proposition 3 weights points by $1/\sigma^2$, a handful of such points dominated and the required sample size collapsed to zero — that is, the analysis reported that the models were trivially separable. The binomial model is also inapplicable on its own terms here: $N_{\text{ex}} \geq N_{\text{tot}}$ on 4.9% of rows, where the binomial variance is zero or negative. Under the Poisson log-ratio the weight distribution is unremarkable — the mean of $1/\sigma^2$ is 3.0 times its median — and the requirement becomes meaningful. We do not quote a magnitude for the binomial inflation: that model was replaced before the separation experiment was first committed, so the figure cannot be regenerated from released code (Appendix F). Counting noise is a lower bound on total noise, so every reported requirement is a lower bound.

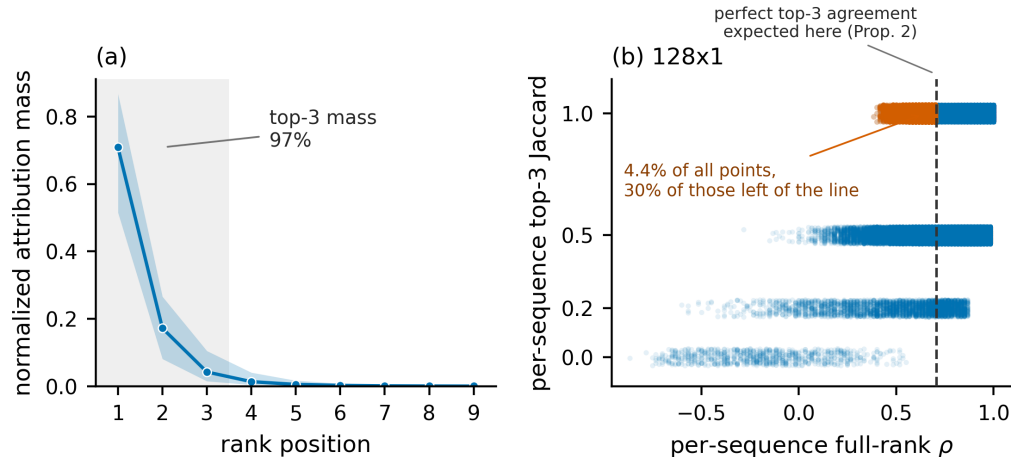


Figure 4: **A low rank correlation has two incompatible readings, and the top- k set separates them.** Accuracy-tied independently trained pairs at width 128, depth 1; three seeds, 203,200 (sequence, pair) points. (a) Normalized attribution mass by rank position, median over 24,000 held-out sequence profiles with interquartile band: a median 97% of the mass sits in the top three positions and the remainder collapses to near zero. (b) Per-sequence full-rank ρ against per-sequence top-3 set Jaccard. The dashed line is $\rho = 0.708$, the value Proposition 1 gives for two models that agree *exactly* on the top 3 and order the remaining 6 at random. Points left of it with Jaccard = 1 (high-lighted) are cases where the two models pick the same three positions while the rank correlation reads as disagreement. **There are 8,861 such (sequence, pair) observations: 4.4% of all 203,200 plotted, and 30% of the 29,310 that fall left of the line.** Each observation is one held-out sequence under one accuracy-tied pair, so the region is a substantial population rather than a handful of outliers. The statistic cannot distinguish the two readings; the top- k set can.

check	what it reported	what it actually tested	corrected control
Ceiling control	Held-out $R^2 = 1.000000$ recovering a target the architecture realizes exactly, in every cell and every seed.	That Adam does not leave a perfect solution. Warm-started <i>at</i> the target, its initial loss was already below the stopping threshold and best-iterate selection kept the initial state, so the loop never ran.	Cold-started on the same target, the worst cell reaches only 0.4501 and 11 of 12 cells fall below the 0.999 threshold.
Noise model	A required sample size of zero: any two fits separable with no data at all.	Inverse-variance weighting under a count model undefined on this library: the pools are not a proportion, $\text{ex} \geq \text{tot}$ on 4.9% of rows, so binomial variance is zero or negative there and a few near-saturated sequences carried the statistic.	Under the Poisson log-ratio the weights are ordinary (mean $1/\sigma^2$ is $3.0\times$ its median) and the requirement is finite, 0.5 to 87,751,464 measurements.

Table 3: **Both controls we relied on were passing for reasons unrelated to what they were meant to test.** Neither failure was visible from the reported number alone: a control that cannot fail and a weighting scheme that silently concentrates on a few points both look like success.

174 4.4 Cause 4: two ways our validation nearly hid the failure

175 Both are errors we committed and caught. We report them because the controls that catch them are
 176 cheap and appear not to be standard, in the spirit of the randomization checks that exposed saliency
 177 methods insensitive to their own model [1]. Table 3 states each one as what was reported, what was
 178 actually tested, and what the corrected check reports.

179 **A ceiling that was arithmetic, not evidence.** To check that a refit can recover a target, one refits
 180 to a target the class provably contains. Ours warm-started from the reference weights, so for the
 181 null target — the reference’s own latent — the initial loss was already 6.7×10^{-19} , below the
 182 early-stopping threshold; best-iterate selection kept the initial state and the loop never ran. The
 183 control returned 1.000000 because it *began at the answer*. It verified that the optimizer does not
 184 wander away from a perfect solution, never that the procedure can *find* one it does not start at — the
 185 capability every downstream conclusion assumed (Appendix D).

186 The fix is a ceiling that requires search: cold-start the same refit. Under that control, recovery of a
 187 target the architecture realizes *exactly* falls to held-out $R^2 \approx 0.45$ at depth 3, with per-seed variation
 188 indicating distinct basins rather than measurement noise.

189 **A negative reported from an empty search.** A separate experiment reported “no witness found”
 190 after examining zero candidates: every cell it searched had an empty candidate set for structural
 191 reasons. Searched-and-found-nothing is a result; never-searched is not, and the second was reported
 192 as the first. We now emit a VOID verdict whenever a negative would be drawn from an empty
 193 candidate set, with a regression test pinning the distinction.

194 4.5 Boundary conditions

195 We found no regime in this grid where the practice is safe. What we can give is the shape of the
 196 degradation, which is measurable and consistent.

- 197 • **Depth is what tracks the failure; width is not, and no cell is safe.** Top-3 divergence
 198 rises monotonically with depth: 43–74% at depth 1, 73–88% at depth 2, 90–94% at depth
 199 3, a rank correlation of +0.80 with depth against −0.06 with width. Within depth 1 it
 200 improves with width, from 74% at width 16 to 43% at width 128, but it never falls below
 201 43% anywhere in the grid. We looked for a low-capacity exemption and found the opposite:
 202 the narrowest models are among the worst.
- 203 • **Fit quality predicts agreement but does not explain it.** The correlation between held-out
 204 R^2 and agreement is +0.650: real, but far from explanatory, and the best-fitting cell still
 205 shows the failure.
- 206 • **Pairs that are harder to tell apart agree more, not less.** Across cells the median number
 207 of held-out measurements needed to separate a tied pair as functions is rank-correlated
 208 −0.94 with top-3 divergence: where the two fits are closer as functions they also pick more
 209 similar positions, which is the direction one would want and the opposite of what the twin
 210 comparison shows (Appendix 5). The failure is therefore not a by-product of the two fits
 211 being nearly identical.
- 212 • **Scope.** One assay, one phenotype, sequence length 9. We do not claim this generalizes to
 213 longer contexts or to models with strong architectural priors, and we tested only in-silico
 214 mutagenesis, not gradient or Shapley-based attribution.

215 4.6 Actionable takeaways

- 216 1. **Report per-instance top- k sets, not a library-averaged rank correlation.** By Proposi-
 217 tion 1 the averaged correlation cannot distinguish perfect agreement from substantial dis-
 218 agreement.
- 219 2. **Report the concentration of the attribution.** Without it a rank correlation cannot be
 220 interpreted at all: the reference value under perfect agreement depends on it.
- 221 3. **Derive the noise scale from the assay where counts are available,** and check the count
 222 model. Ours was wrong in the direction that made the problem disappear.
- 223 4. **Train more than one seed and report top- k set agreement across them.** A constant-
 224 factor cost, and the cheapest check that catches this.
- 225 5. **Verify that a passing control required a search.** A control that starts at its own target
 226 passes by arithmetic.
- 227 6. **Refuse to report a negative from an empty candidate set.**

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277 A Formal statements

278 The statements below are elementary. We give them explicitly because two of them (Propositions 1
279 and 2) determine how the main-text numbers should be read, and because the failure described in
280 Section 4.3 was a violation of Proposition 2 that produced a plausible-looking wrong answer rather
281 than an error.

282 **Remark 1** (Predictive risk does not determine the attribution). *Let $\mathcal{F}$ be a model class, R a popu-*
283 *lation risk, and $\mathcal{F}_\epsilon = \{f \in \mathcal{F} : R(f) \leq \min_{f'} R(f') + \epsilon\}$ the ϵ -Rashomon set. Let $A : \mathcal{F} \rightarrow \mathbb{R}^L$*
284 *be the attribution map of Eq. (1). Nothing in the definition of $\mathcal{F}_\epsilon$ constrains A to be constant on it.*
285 *Consequently, the attribution reported by a fitting procedure is a function of which element of $\mathcal{F}_\epsilon$*
286 *the optimizer returns, and that is determined by the initialization, not by the data. The empirical*
287 *content of this paper is the diameter of $A(\mathcal{F}_\epsilon)$ on a real assay, measured at the level of the top- k set.*

288 **Proposition 1** (Expected rank correlation under concentrated attribution). *Let two models rank L*
289 *positions. Suppose they agree exactly on the ordering of the top k positions, and that the remaining*
290 *$m = L - k$ positions carry negligible attribution mass, so that their relative orderings are inde-*
291 *pendent uniform random permutations. Then the expected Spearman rank correlation over all L*
292 *positions is*

$$\mathbb{E}[\rho_L] = 1 - \frac{m(m^2 - 1)}{L(L^2 - 1)}. \quad (2)$$

293 For $L = 9$ and $k = 3$ this is $1 - 2^{10}/720 = 0.708$.

294 *Proof.* Spearman's ρ over L items is $\rho = 1 - 6 \sum_i d_i^2 / (L(L^2 - 1))$, where d_i is the rank difference
295 at item i . The top k contribute $d_i = 0$ by assumption. For two independent uniform permutations of
296 m items, $\mathbb{E}[\rho_m] = 0$, and since $\rho_m = 1 - 6 \sum d_i^2 / (m(m^2 - 1))$ this gives $\mathbb{E}[\sum d_i^2] = m(m^2 - 1)/6$
297 over the tail block. Substituting yields the result. $\square$

top- k agreed	$k = 2$	$k = 3$	$k = 4$	$k = 5$
$\mathbb{E}[\rho_9]$	0.533	0.708	0.833	0.917

Table 4: Expected Spearman correlation over $L = 9$ positions when two models agree *exactly* on the top k and order the rest at random (Proposition 1). A reported ρ at or below these values carries no evidence of disagreement about which positions matter.

298 **Proposition 2** (Poisson log-ratio noise scale). *Let the phenotype be $y = a \log_{10}(N_{\text{ex}}/N_{\text{tot}}) + b$ with*
299 *N_{ex} and N_{tot} independent Poisson counts with means $\lambda_{\text{ex}}, \lambda_{\text{tot}}$. Then to first order*

$$\text{sd}(y) = \frac{|a|}{\ln 10} \sqrt{\frac{1}{\lambda_{\text{ex}}} + \frac{1}{\lambda_{\text{tot}}}} \approx \frac{|a|}{\ln 10} \sqrt{\frac{1}{N_{\text{ex}}} + \frac{1}{N_{\text{tot}}}}. \quad (3)$$

300 *Proof.* For $N \sim \text{Poisson}(\lambda)$, the delta method with $g(N) = \ln N$ gives $\text{Var}(\ln N) \approx$
301 $\lambda^{-2} \text{Var}(N) = 1/\lambda$. Independence of the two pools gives $\text{Var}(\ln N_{\text{ex}} - \ln N_{\text{tot}}) \approx 1/\lambda_{\text{ex}} + 1/\lambda_{\text{tot}}$.
302 Converting to base 10 and scaling by a yields the result; substituting counts for means is the usual
303 plug-in estimator. $\square$

304 **Remark 2** (Why the binomial form fails here, and fails silently). *Modelling the readout as a propor-*
305 *tion p from N trials gives $\text{sd}(y) \propto \sqrt{(1-p)/(Np)}$, which tends to 0 as $p \rightarrow 1$. Since the statistic*
306 *of Proposition 3 weights points by $1/\sigma_i^2$, a small number of near-saturated sequences then carry*
307 *unbounded weight. Under Proposition 2 the ratio $\text{mean}(1/\sigma^2)/\text{median}(1/\sigma^2)$ is 3.0, a heavy but*
308 *unremarkable tail. We give no corresponding figure for the binomial form: it was replaced before*
309 *the separation experiment was first committed and cannot be regenerated from released code, so*
310 *only the structural defect is quoted. That defect is decisive on its own: $N_{\text{ex}} > N_{\text{tot}}$ in 4.9% of rows,*
311 *by up to $20.9\times$, so the two counts are not numerator and denominator of a proportion.*

312 **Proposition 3** (Separation sample size). *Let f_1, f_2 be two candidate functions with pointwise differ-*
313 *ence $d_i = f_1(x_i) - f_2(x_i)$ at held-out points x_i , and let observations carry independent Gaussian*

noise with known standard deviations σ_i . Testing $H_0 : y \sim f_1$ against $H_1 : y \sim f_2$ at two-sided level α with power $1 - \beta$ requires

$$n \geq \frac{(z_{1-\alpha/2} + z_{1-\beta})^2}{(d_i/\sigma_i)^2}, \quad \overline{(d_i/\sigma_i)^2} = \frac{1}{n} \sum_{i=1}^n \frac{d_i^2}{\sigma_i^2}. \quad (4)$$

Proof. The log-likelihood ratio between the two simple hypotheses is Gaussian with mean $\pm \frac{1}{2} \sum_i d_i^2/\sigma_i^2$ under H_0 and H_1 respectively, and variance $\sum_i d_i^2/\sigma_i^2$. The standardized separation is therefore $(\sum_i d_i^2/\sigma_i^2)^{1/2}$, and the standard two-sided power requirement $(\sum_i d_i^2/\sigma_i^2)^{1/2} \geq z_{1-\alpha/2} + z_{1-\beta}$ rearranges to the stated bound. $\square$

Remark 3 (The reported requirement is a lower bound). *Proposition 2 accounts for sequencing count noise only. Library preparation, transfection and biological variation add further variance, which increases σ_i and hence increases the required n . Every separation requirement in the main text is therefore a lower bound on the measurements needed.*

Proposition 4 (Effective number of contributing positions). *For an attribution vector $A \in \mathbb{R}_{\geq 0}^L$ with $\sum_j A_j > 0$, let $q_j = A_j / \sum_{j'} A_{j'}$. The effective number of contributing positions is $N_{\text{eff}}(A) = \exp(-\sum_j q_j \ln q_j)$, satisfying $1 \leq N_{\text{eff}} \leq L$, with $N_{\text{eff}} = L$ iff all positions contribute equally and $N_{\text{eff}} \rightarrow 1$ as one position dominates.*

This is the exponentiated Shannon entropy of the normalized attribution and is reported per-sequence in Table 2; the median over held-out sequences is 3.92 of 9 for accuracy-tied independently trained pairs and 4.38 for the twin comparison, both medians *across cells*; per cell the value ranges from 2.20 to 7.80, so the reference value in Proposition 1 must be read against the cell in question rather than against the grid median.

B Full per-cell results

Table 5 combines Tables 1 and 2 for every cell in the width $\times$ depth grid, with bootstrap intervals on the columns that admit them. It is generated by `experiments/make_tables.py` directly from the committed run directories, as are all tables in this paper.

cell	held-out R^2	tied pairs	res.	ρ p10 / med	$\rho < 0.5$	top-3 mass	eff. pos	top-1 diff	exact	Jaccard
16 $\times$ 1	0.6011 [0.5725, 0.6274]	162 [156, 169]	0.0699	+0.217 / +0.650	30%	0.926	3.03	45%	26%	0.57
16 $\times$ 2	0.5591 [0.5243, 0.5880]	158 [147, 167]	0.1059	-0.167 / +0.417	58%	0.819	4.44	60%	13%	0.43
16 $\times$ 3	0.5349 [0.4948, 0.5677]	134 [122, 147]	0.1161	-0.300 / +0.283	69%	0.768	5.11	68%	9%	0.37
32 $\times$ 1	0.6019 [0.5754, 0.6263]	167 [160, 172]	0.0651	+0.365 / +0.717	19%	0.956	2.56	37%	32%	0.62
32 $\times$ 2	0.5537 [0.5110, 0.5909]	143 [128, 158]	0.1032	-0.168 / +0.408	58%	0.851	4.05	62%	12%	0.42
32 $\times$ 3	0.4792 [0.4128, 0.5349]	102 [97, 108]	0.1256	-0.317 / +0.258	71%	0.759	5.28	69%	8%	0.36
64 $\times$ 1	0.6278 [0.6003, 0.6544]	160 [151, 169]	0.0402	+0.600 / +0.833	4%	0.969	2.36	24%	48%	0.73
64 $\times$ 2	0.5381 [0.4846, 0.5861]	100 [81, 120]	0.0882	-0.183 / +0.417	57%	0.873	3.79	61%	14%	0.43
64 $\times$ 3	0.3929 [0.3309, 0.4483]	92 [86, 97]	0.1267	-0.383 / +0.175	78%	0.540	7.80	74%	6%	0.31
128 $\times$ 1	0.6372 [0.6122, 0.6585]	154 [138, 166]	0.0318	+0.700 / +0.883	2%	0.976	2.20	18%	57%	0.78
128 $\times$ 2	0.5836 [0.5369, 0.6259]	144 [119, 168]	0.0461	+0.400 / +0.750	16%	0.918	3.16	36%	27%	0.60
128 $\times$ 3	0.4761 [0.3949, 0.5456]	95 [71, 117]	0.0937	-0.267 / +0.317	67%	0.728	5.67	68%	10%	0.38

Table 5: **No cell is withheld: the pattern in Tables 1 and 2 holds across the whole grid.** All twelve cells, combining both main-text tables. **Which columns carry intervals, and why the rest do not:** held-out R^2 and the tied-pair count are per-seed quantities and are given as mean with a 95% bootstrap interval over the ten seeds. The remaining columns are medians pooled over every tied pair across all seeds rather than per-seed values, so a seed-level bootstrap interval is not defined for them and none is invented here.

C Protocol and pre-registration

Table 6 gives the per-cell protocol, read from the `tuning_budget.json` written with each run and from the per-seed records. The split rule is the same everywhere: for seed k a generator seeded with $1,000,003 + k$ draws a permutation of the 4000 subsampled sequences, the first 3000 form the fit split and the remaining 1000 are held out, and the indices are recorded in the run directory. All twelve cells use ten seeds.

cell	learning rate chosen (seeds)	configs	epochs	grad. steps
16×1	0.01	4	1,200	1,200
16×2	0.01	4	1,200	1,200
16×3	0.01 (×9), 0.003 (×1)	4	1,200	1,200
32×1	0.01	4	1,200	1,200
32×2	0.01	4	1,200	1,200
32×3	0.01	4	1,200	1,200
64×1	0.01 (×4), 0.003 (×6)	4	1,200	1,200
64×2	0.01 (×9), 0.003 (×1)	4	1,200	1,200
64×3	0.01 (×9), 0.003 (×1)	4	1,200	1,200
128×1	0.003 (×9), 0.001 (×1)	4	1,200	1,200
128×2	0.003 (×4), 0.001 (×6)	4	1,200	1,200
128×3	0.003 (×4), 0.001 (×6)	4	1,200	1,200

Table 6: **The tuning budget is four learning rates per cell and nothing else, so the disagreement cannot be an artifact of unequal search.** Per-cell protocol, read from `tuning_budget.json` and the per-seed records. The search space is “learning rate over [0.01, 0.003, 0.001, 0.0003], selected once per cell and shared by all models in it”. The selection rule is identical in every cell — “no model selection: every trained model is kept and enters the pairwise analysis” — so it is stated here rather than repeated as a column. **The chosen rate is not constant across seeds:** it varies in 7 of 12 cells, and the multiplicities are given. Within any one seed a single rate is shared by all 20 models in the cell, so the models still differ only in initialization; across seeds the rate is re-selected.

343 Claims, thresholds and go/no-go criteria were written and committed before any result existed:
344 PREREGISTRATION.md has a single commit, dated 2026-08-26, and the earliest committed run di-
345 rectory is dated 2026-08-29, so the ordering is checkable from the repository history rather than
346 asserted. Amendment is additionally blocked by a pre-write hook in the authoring environment
347 that matches the target path of a write; we name the mechanism precisely because it constrains our
348 workflow rather than the repository, and the durable evidence is the single-commit history. Every
349 reported number carries the git SHA and configuration hash of the run that produced it.

350 D The vacuous ceiling, in full

351 The refit loads the reference weights, then evaluates the loss and snapshots the best iterate *before*
352 entering the optimization loop. For the null target — the reference’s own latent — that initial state
353 is already the best iterate that will ever be seen, so no later step can displace it. The measured
354 initial loss is 6.7×10^{-19} , below the 10^{-10} early-stopping threshold, and the loop breaks on its first
355 iteration.

356 The consequence is visible in the run records without reference to any loss value: the warm-started
357 control returns held-out R^2 of exactly 1.0 in all 120 cell–seed combinations, bit-exact rather than
358 approximate. A control that cannot return anything else has not measured anything. The cold-started
359 replacement, which must actually travel to the same in-class target, recovers it at only 0.4501 in the
360 worst cell and falls below the 0.999 threshold in 11 of 12 cells, with per-seed spread of 0.21 to 0.87
361 at width 128 depth 3. Both arms are plotted in Figure 6.

362 The general form of the error is worth stating separately from its instance. A control that verifies a
363 procedure can recover a target must ensure the procedure does not *begin* at that target. Warm-starting
364 a refit from the reference weights and then asking it to recover the reference’s own output satisfies
365 the letter of the control and none of its purpose.

366 E Reproducibility

367 Code, pre-registered analysis plans, per-seed raw values, and run provenance are released at
368 <https://github.com/shadi97kh/CERTGNN>. Each result directory records the git SHA, the con-
369 figuration hash, the seed list, and the tuning budget consumed, and every table and figure in this
370 paper is generated from those directories by `experiments/make_tables.py`, `make_figures.py`
371 and `make_workflow_figure.py` rather than transcribed.

F Noise model derivation

Z requires an observation-noise scale. This library supplies one rather than needing an assumption, and the derivation is given here because the choice of count model changes Z by orders of magnitude.

The phenotype is affine in a log count ratio. Each sequence carries two count pools, ex and tot . Regressing the phenotype y on $\log_{10}(ex/tot)$ over the full library gives slope 0.9331 [0.9314, 0.9345] across seeds, and $r = 0.990$ over the full 30,483-sequence library. The relationship is tight enough that the count ratio, not the phenotype, is the quantity whose noise must be modelled.

Why Poisson and not binomial. The two pools are *not* numerator and denominator of a proportion. Over the committed library, ex exceeds tot in 4.9% of rows, with ratios as large as 20.9, and $ex = 0$ in 11.4% of rows. A binomial model is therefore not merely inefficient but undefined on roughly one row in twenty: $p = ex/tot \geq 1$ there, so the binomial variance $p(1-p)/tot$ is zero or negative.

Delta method. Treating both pools as independent Poisson variates and applying the delta method to the log of their ratio,

$$\text{var}(\log_{10} R) = \frac{1}{\ln(10)^2} \left(\frac{1}{ex} + \frac{1}{tot} \right), \quad \text{sd}(y_i) = |a| \frac{\sqrt{1/ex_i + 1/tot_i}}{\ln 10}, \quad (5)$$

with a the fitted slope and a half-count continuity correction applied to both pools, which also covers the rows with $ex = 0$. In the standardized units the models see, the median of this quantity is 0.3114 [0.3051, 0.3194].

The diagnostic that surfaced the binomial failure. The separation test weights points by $1/\sigma_i^2$, so a point assigned near-zero noise dominates the statistic. Under a binomial model a near-saturated sequence receives exactly that. The symptom was a weight distribution with a mean far above its median and a required sample size that collapsed to zero for every cell — an implausible answer that is what prompted the check. Under the committed Poisson log-ratio model the mean of $1/\sigma^2$ is 3.0 times its median, a heavy but unremarkable tail, and Z becomes finite and cell-dependent. An unexamined count model would have reported that no data at all is needed to separate the fits.

We deliberately do not quote a magnitude for the binomial inflation. That model was replaced before the first commit of the separation experiment, so its exact variance form is not in the repository and the figure cannot be regenerated from committed code. What *is* reproducible from the committed library is the structural defect above — $p \geq 1$ on 4.9% of rows, hence zero or negative binomial variance — together with the Poisson ratio of 3.0. Both were re-derived from the committed 30,483-row library while preparing this appendix and agree with the recorded values.

A figure comparing the two weight distributions was considered and cut. It would need the binomial variance form, which is not in the repository, and per-observation $1/\sigma^2$ values, which no run directory stores — only the median noise scale survives into `results.json`. Drawing it would have meant inventing the comparator and recomputing the rest outside any recorded run, so the comparison stays in prose where its provenance can be stated.

Z is a lower bound. This models sequencing noise only. Library preparation and biological variation add more, so the true noise is larger and the required sample size larger still. Every Z in this paper should be read as “at least this many”.

G Separability against a reparameterized twin

The main text measures *independently trained* models that predictive accuracy cannot separate, and shows that the assay itself separates them cheaply — fewer than three held-out measurements. This section reports the complementary case: a single fit against a *reparameterized twin* of itself, where the data genuinely cannot adjudicate.

The two cases bracket the problem. In the main-text population the divergence is avoidable: one measurement would settle which of the two models is right, and the selection procedure simply does not ask. Here it is not avoidable — at width 128 depth 3 the twin needs 8.8×10^7 measurements, more than three orders of magnitude beyond the full library — and the top-3 sets diverge regardless. Attribution therefore fails in both regimes, for different reasons, and only the first has a cheap remedy.

The two sets of numbers are not interchangeable. The twin comparison finds substantially more top-3 agreement than independently trained pairs do, so quoting a twin figure for the main-text population would understate the failure.

Run 20260830T030657Z_972496a_5da8cb38, git 972496a, config 5da8cb38, 10 seeds, sequence length 9.

width	depth	held-out R^2	per-instance median ρ	Z to separate
16	1	0.984087 [0.979785, 0.987518]	0.892 [0.877, 0.903]	1 [0, 1]
16	2	0.999241 [0.998708, 0.999619]	0.978 [0.973, 0.983]	42 [25, 60]
16	3	0.999583 [0.999138, 0.999916]	0.984 [0.978, 0.990]	313 [87, 571]
32	1	0.992155 [0.990567, 0.993674]	0.872 [0.847, 0.893]	0 [0, 1]
32	2	0.997328 [0.995451, 0.999059]	0.915 [0.892, 0.937]	87 [14, 193]
32	3	0.999166 [0.998775, 0.999529]	0.923 [0.898, 0.945]	833 [18, 2,136]
64	1	0.998324 [0.997555, 0.998958]	0.899 [0.886, 0.913]	2 [1, 3]
64	2	0.999997 [0.999995, 0.999998]	0.855 [0.832, 0.880]	4,176 [1,320, 7,333]
64	3	0.999996 [0.999993, 0.999998]	0.847 [0.835, 0.860]	7,300 [2,156, 14,436]
128	1	0.999876 [0.999862, 0.999889]	0.923 [0.913, 0.933]	9 [7, 11]
128	2	1.000000 [1.000000, 1.000000]	0.905 [0.888, 0.920]	3,158,529 [1,402,633, 5,004,083]
128	3	1.000000 [1.000000, 1.000000]	0.890 [0.885, 0.895]	87,751,464 [39,140,215, 138,392,840]

Table 7: **A fit and its own twin are far closer than two independently trained fits, which is why the two populations must not be quoted interchangeably.** Z is the number of held-out measurements needed to reject that the two are the same function at $\alpha = 0.05$ with power 0.8, under the noise model of Appendix F. It is a lower bound: it counts sequencing noise only.

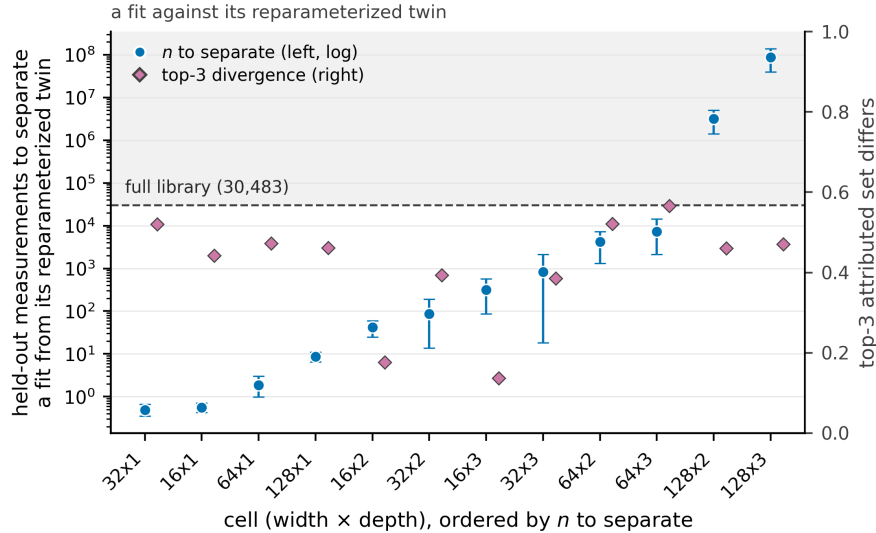


Figure 5: **Separability and top-3 divergence for a fit against its own reparameterized twin, not for independently trained pairs.** The requirement spans nine orders of magnitude here and two cells exceed the full library, where the main-text population separates with fewer than three measurements in every cell (Figure 3). Left axis logarithmic; the dashed line is the full library.

H Cold-start recovery, per cell

The defect. The held-out closure experiment (run 20260829T181937Z_b2951ae_9fdd4067, git b2951ae) reported that no cell contains the reparameterized twin, on the strength of a ceiling control that does not do the work it appears to do. Its refit path, `closure_heldout.refit_on_split`, loads the reference weights, then evaluates the loss and snapshots the best iterate *before* entering the optimisation loop. For the null target — which is the reference’s own latent — the refit therefore begins at the answer. The measured initial loss is 6.7×10^{-19} , already below the 10^{-10} early-stop tolerance, so the loop breaks on its first iteration and returns the reference unchanged.

A held-out R^2 of 1.000000 on that target is arithmetic, not evidence. It establishes that Adam does not wander away from a perfect solution. It does not establish that the refit can *find* a solution it does not start at, which is the capability the containment conclusion depends on. Best-iterate selection is what makes the failure silent: because the initial state is already the best iterate ever seen, no subsequent step can displace it, and the procedure reports a perfect score without having searched.

The control that replaced it. Run 20260829T191427Z_94dd5a2_2267b74e, git 94dd5a2, config 2267b74e, 10 seeds. The null target is refit *without* warm-starting from the reference, from a cold random initialization. The target is known to be in the class, since the reference realises it, so this asks only whether the optimiser can travel to it. Mean initial loss across cells confirms the refit no longer begins at the answer.

width	depth	cold-start null, held-out R^2	warm warp refit, held-out R^2
16	1	0.9988 [0.9982, 0.9993]	0.9748 [0.9669, 0.9809]
16	2	0.9476 [0.9244, 0.9703]	0.9935 [0.9895, 0.9962]
16	3	0.7763 [0.6820, 0.8589]	0.9959 [0.9921, 0.9986]
32	1	0.9994 [0.9990, 0.9997]	0.9749 [0.9708, 0.9789]
32	2	0.8258 [0.7303, 0.9103]	0.9463 [0.9185, 0.9724]
32	3	0.5626 [0.4404, 0.6891]	0.9404 [0.9151, 0.9643]
64	1	0.9971 [0.9955, 0.9984]	0.9804 [0.9696, 0.9893]
64	2	0.7122 [0.5620, 0.8561]	0.9236 [0.8929, 0.9532]
64	3	0.4501 [0.3529, 0.5575]	0.9034 [0.8879, 0.9186]
128	1	0.9969 [0.9962, 0.9975]	0.9885 [0.9868, 0.9900]
128	2	0.8997 [0.8316, 0.9529]	0.9575 [0.9381, 0.9734]
128	3	0.5328 [0.4000, 0.6714]	0.9247 [0.9108, 0.9394]

Table 8: The ceiling that requires search. The cold-start column is an upper bound on what this procedure can say about any target, because its target is in the class by construction. Mean cold-start initial loss ranges from 1.826 to 2.093, against 6.7×10^{-19} for the warm-started control it replaces.

What the control shows. Read by depth: at depth 1 the worst-cell cold-start null reaches 0.9969 against 0.9748 for the warm warp refit, so the ceiling sits above the measurement; at depth 2 the worst-cell cold-start null reaches 0.7122 against 0.9236 for the warm warp refit, so the ceiling sits below the measurement; at depth 3 the worst-cell cold-start null reaches 0.4501 against 0.9034 for the warm warp refit, so the ceiling sits below the measurement.

At depth 1 the ceiling sits above the measurement, so the comparison is valid there and the warp target is genuinely not reached as well as an in-class one. At depths 2 and 3 the ranking inverts: the warm-started warp refit scores *higher* than a cold-started refit to a target guaranteed to be in the class. That is initialization dominating the result, and no conclusion about what the class contains can be drawn at those depths. The configured pass threshold is 0.999, and 11 of 12 cells fall below it.

The honest reading is therefore narrower than the original claim: the refit procedure did not find a member of the class equal to the twin. That is consistent with no such member existing, and equally consistent with one existing that the optimiser failed to reach.

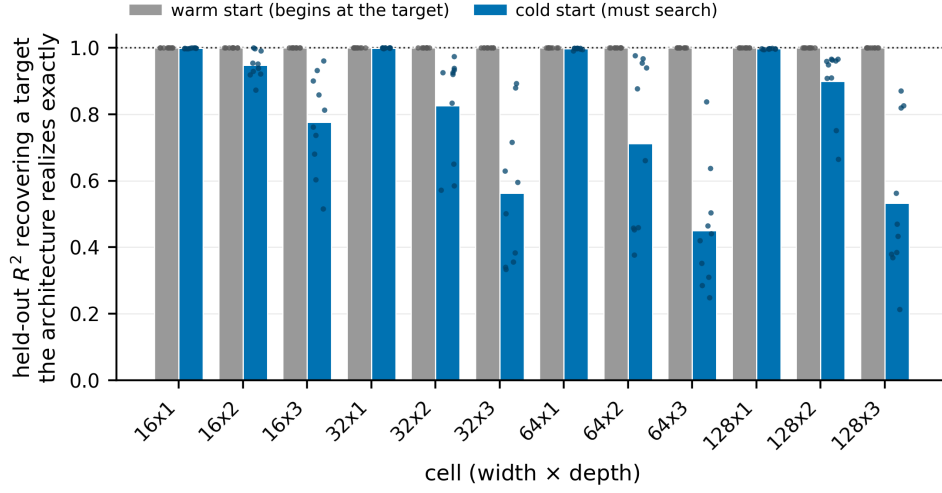


Figure 6: **The warm-start control cannot fail, so it measures nothing.** Held-out R^2 recovering the null target, a function the architecture realizes exactly, with the ten per-seed values overlaid. The warm-started control returns 1.000000 in every cell and every seed — all 120 values are exactly 1.0, not approximately — because it is initialized at the reference and its initial loss of 6.7×10^{-19} is already below the 10^{-10} stopping threshold, so the optimization loop never runs and best-iterate selection keeps the initial state. The cold-started arm must actually search for the same target and its per-seed spread shows how much the basin varies: at 128×3 restarts land between 0.21 and 0.87. Warm bars are flat by construction; that flatness is the finding, not a plotting artefact.